

FLUX

Adaptive Difficulty Survival Game

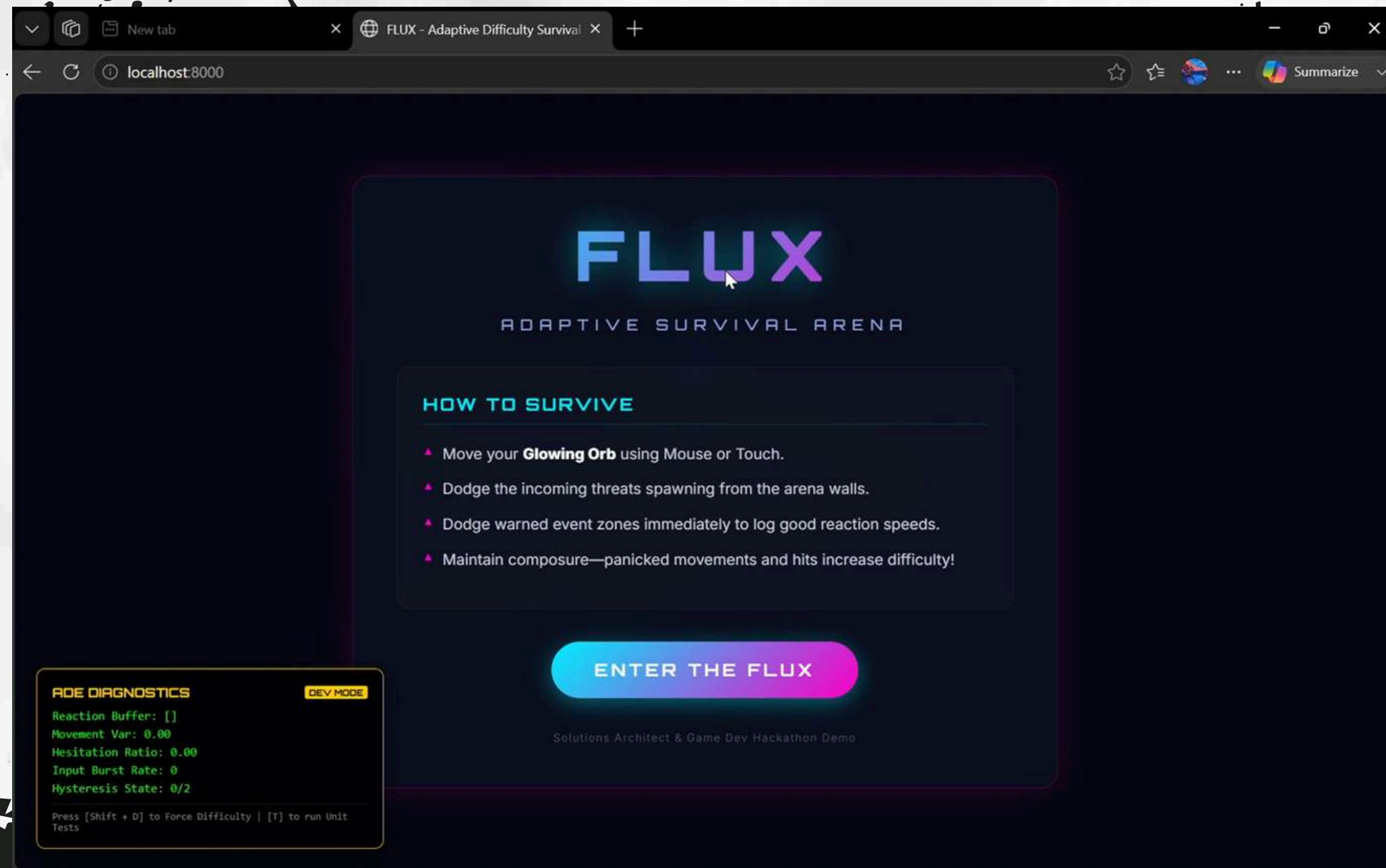
Unlike traditional games where players manually choose difficulty, FLUX automatically moves between tiers based on actual performance.

Neuromorphism in Gaming

Team - LOOPERS

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This is “Flux”



Our project, FLUX, introduces a neuromorphic inspired adaptive gaming system that continuously analyzes player behavior and dynamically adjusts enemy difficulty in real time. Instead of forcing players to adapt to the game, the game adapts to the player, creating a more engaging, personalized, and balanced gaming experience.

Traditional games use fixed difficulty levels such as Easy, Medium, and Hard. However, every player has different skills, reaction times, and play styles.

Existing Problem

- Fixed difficulty settings cannot suit every player.
- Beginners often get frustrated.
- Skilled players become bored quickly.
- Current systems rarely consider real-time player behavior.

Our Solution

FLUX – Adaptive Difficulty Survival Game

A game that continuously monitors:

- Reaction Time
- Movement Patterns
- Player Stress Level
- Near-Miss Events
- Input Frequency and adjusts enemy behavior dynamically.

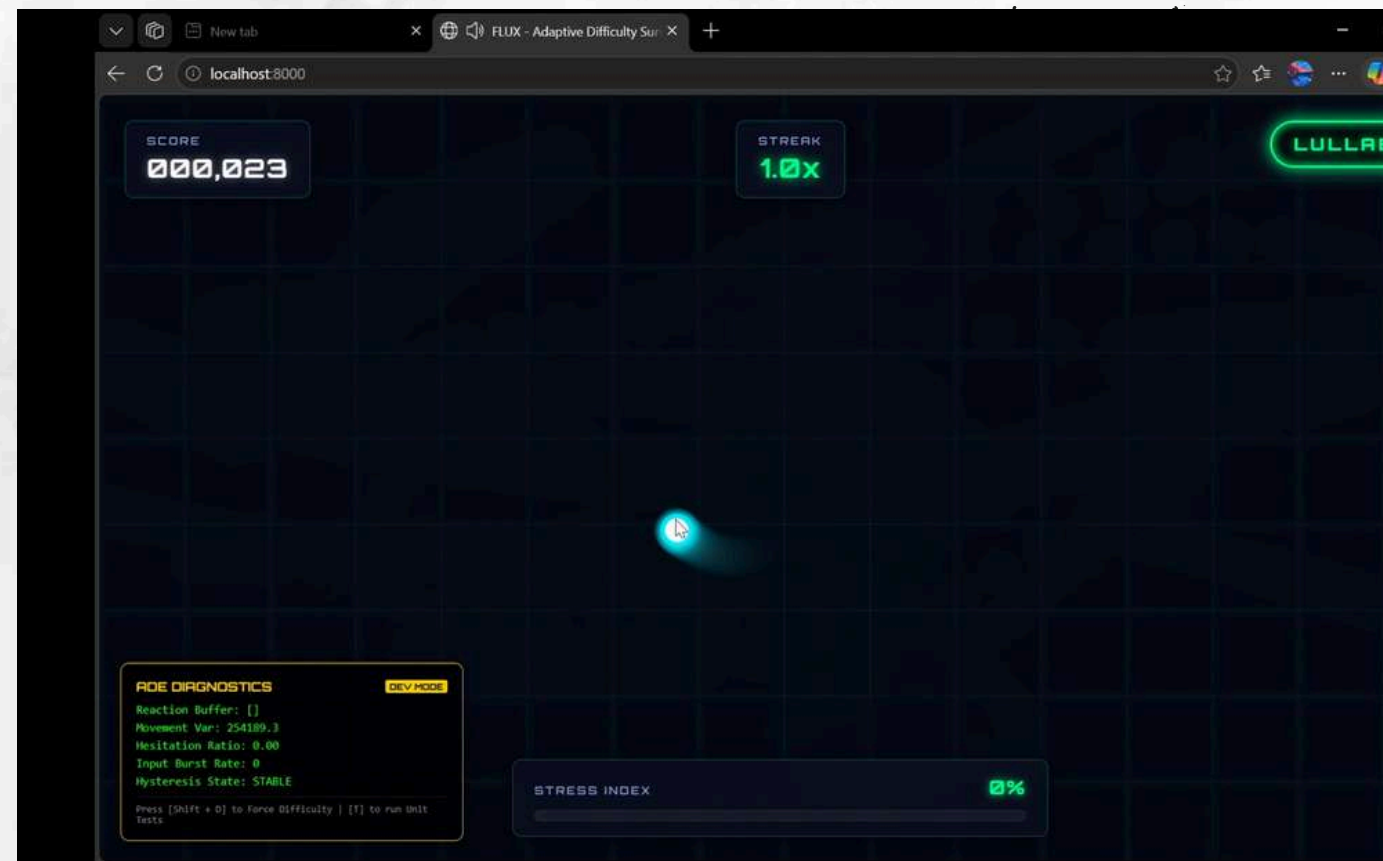
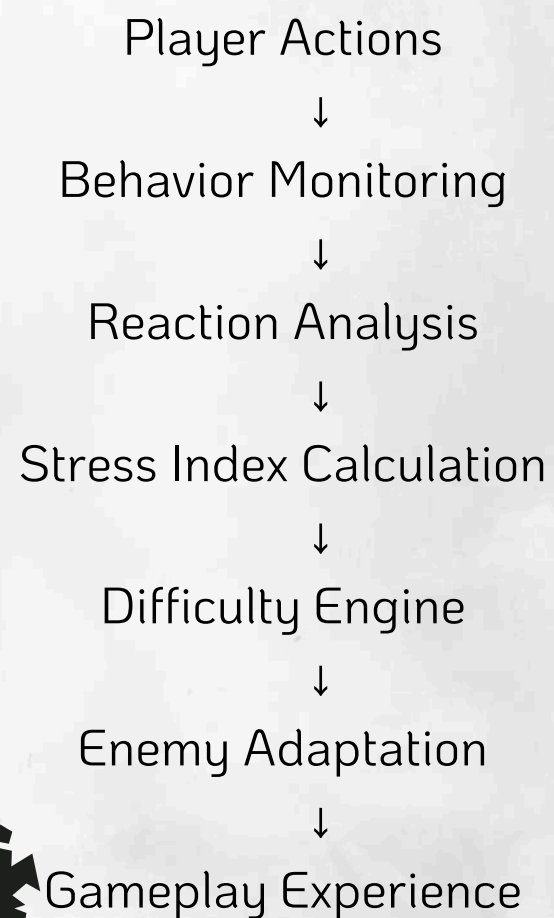
WHAT IS NEUROMORPHISM IN GAMING?

Neuromorphic computing is inspired by how the human brain processes information.

Instead of following rigid rules, the system:

- Observes patterns
- Learns behavior
- Adapts decisions In FLUX,
- Analyze player performance
- Detect skill changes
- Adjust difficulty intelligently

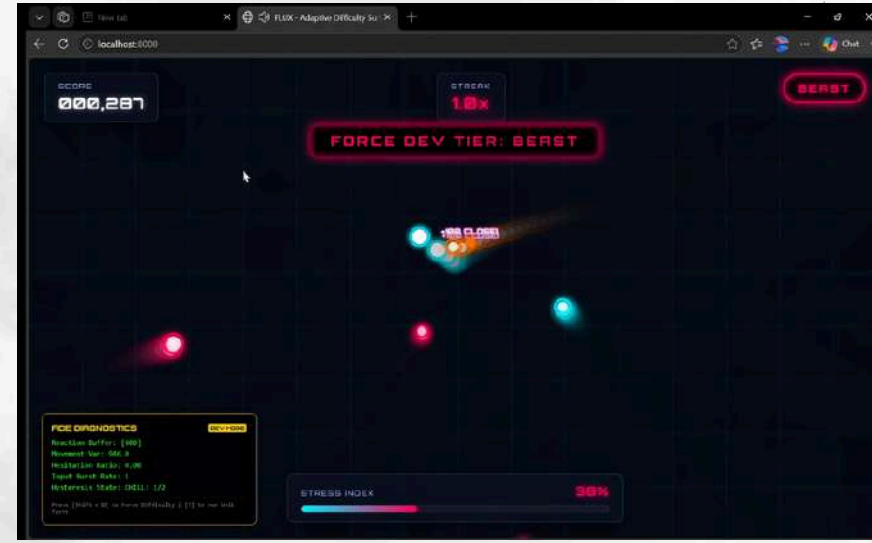
SYSTEM ARCHITECTURE



This is the core gameplay environment.
The player controls the blue orb while avoiding incoming enemy particles.
The system continuously tracks:

- Reaction speed
- Movement smoothness
- Escape efficiency
- Near collision frequency

DIFFICULTY LEVELS



CHILL

- Beginner Friendly
- Slow enemies
- Low pressure

FLOW

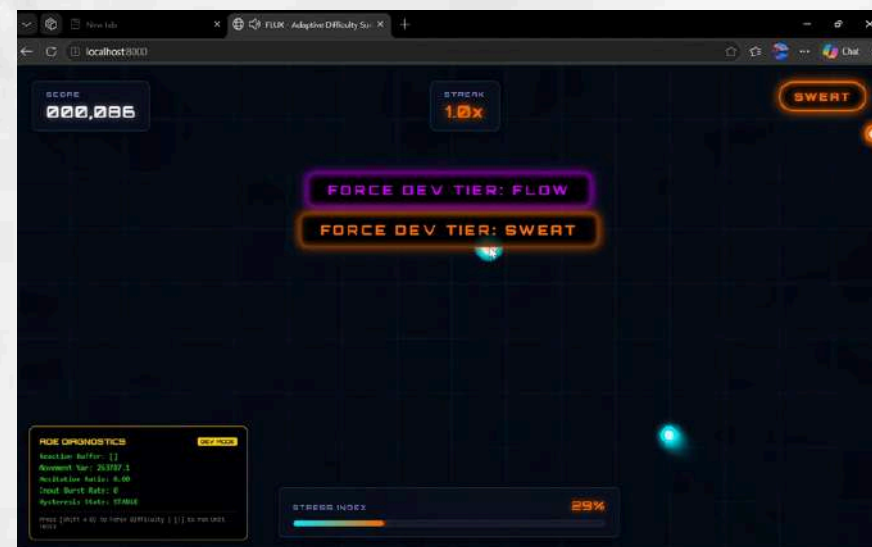
- Balanced experience
- Optimal engagement

SWEAT

- Faster enemy behavior
- Higher challenge

BEAST

- Maximum intensity
- Expert-level gameplay

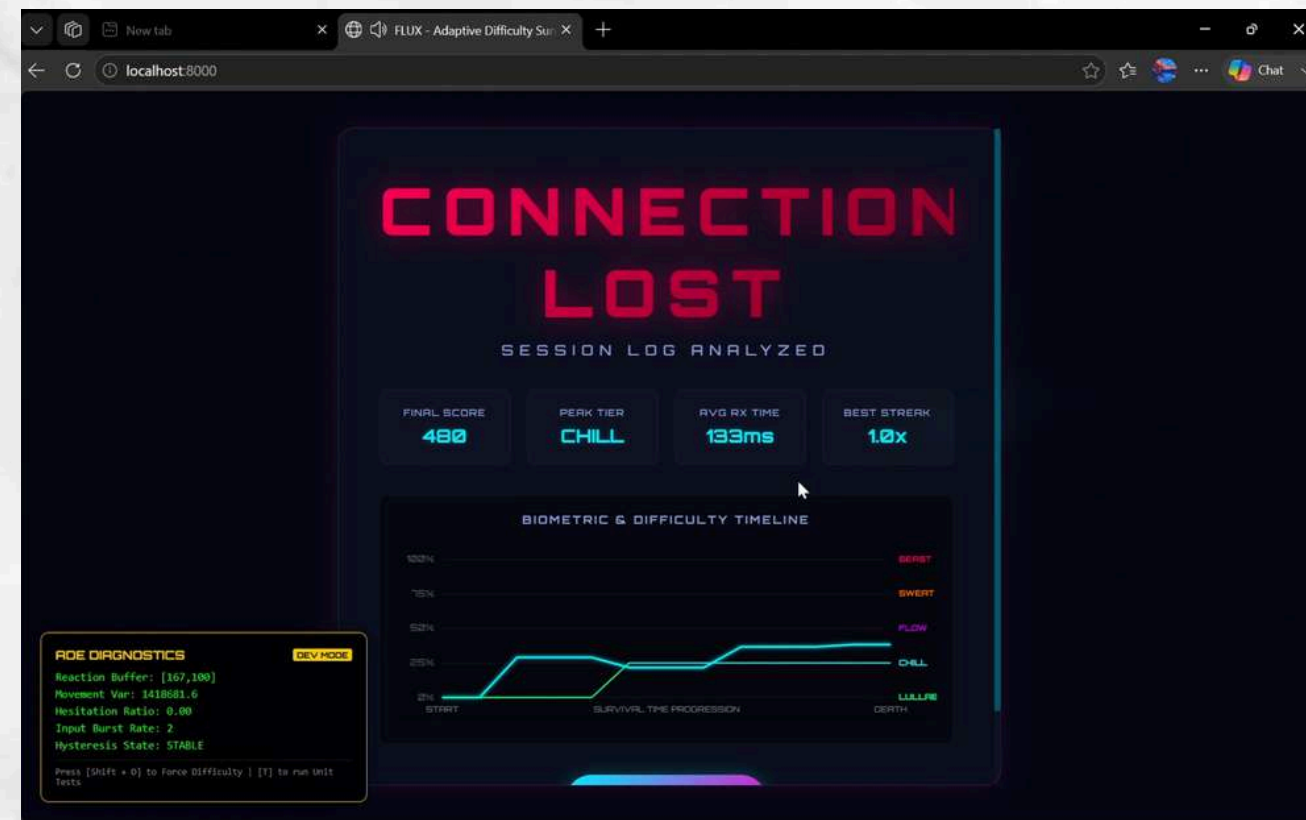


SESSION ANALYTICS

After each session, FLUX generates performance analytics.

Includes:

- Final Score
- Best Streak
- Average Reaction Time
- Peak Difficulty Tier
- Difficulty Timeline



UNIT TESTING

Tests include:

- Stress Index calculations
- Tier stability
- Difficulty transitions
- Collision detection
- Telemetry logging

FUTURE SCOPE

- EEG/Brainwave Integration
- Eye Tracking
- Emotion Detection
- Multiplayer Adaptive Balancing
- Personalized AI Opponents
- VR Adaptive Experiences

CONCLUSION

FLUX demonstrates how neuromorphic inspired intelligence can transform gaming from a one size fits all experience into a personalized adaptive environment. By continuously analyzing player behavior and dynamically adjusting difficulty, the system keeps players engaged, challenged, and immersed while maintaining fairness and enjoyment.

The background is a high-contrast, abstract composition. It features a light gray, textured surface that resembles watercolor or ink on paper. Overlaid on this are various black elements: large, irregular ink splatters and blotches, particularly in the corners; and several thin, dark, brush-like strokes that sweep across the upper and lower portions of the frame. The overall effect is one of raw, expressive energy.

THANK YOU